

Intelligent feature recognition for STEP-NC-compliant manufacturing based on artificial bee colony algorithm and back propagation neural network

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ABSTRACT

This paper presents an intelligent feature recognition method for STEP-NC-compliant manufacturing based on artificial bee colony (ABC) algorithm and back propagation (BP) neural network. In the method, after extracting the geometric and topological information from its STEP AP203 neutral file, the minimum subgraphs of a part are firstly constructed based on the concavity and convexity judgment algorithm. Then, an improved BP neural network used to STEP-NC-compliant manufacturing feature recognition is proposed with the combination with ABC algorithm. Finally, the STEP-NC-compliant manufacturing features in the part are recognized accurately and efficiently after the information data from the minimum subgraphs of the part is input into the improved BP neural network. At the end, it has been concluded by case study that the proposed method is effective and feasible.

1. Introduction

The integration of the CAD/CAPP/CAM/CNC plays an important role in achieving intelligent manufacturing. Aiming at integrating the CAD/CAPP/CAM/CNC, ISO (International Organization for Standardization, ISO) has developed STEP [1,2] (Standard for the Exchange of Product data, STEP) and STEP-NC [3,4] standards. STEP provides a representation of product information along with the necessary mechanisms and definitions to enable product data to be exchanged, while STEP-NC is an extension of STEP that defines a machine-independent, manufacturing features-oriented and bi-directional data standard for CNC. Although these two standards provide a vision for the integration of the CAD/CAPP/CAM/CNC, even for the realization of intelligent manufacturing, STEP defines a standard for CAD design data which can't directly be utilized by downstream CAPP, CAM and CNC which require manufacturing data. Feature recognition can bridge the gap between CAD and CAPP/CAM/CNC. Therefore, feature recognition for STEP-NC-compliant manufacturing becomes a basis of CAD/CAPP/CAM/CNC integration and is one of the key technologies of intelligent manufacturing.

Many scholars have done a lot of important research about feature recognition. For example, Joshi and Chang [5] proposed a graph-based feature recognition method. The method matches feature subgraphs of a

part with pre-defined feature subgraphs to realize the feature recognition. Sunil et al. [6] utilized explicit feature graphs and geometric inference rules to realize the recognition of features. Followed by it, Xie et al. [7] combined the weighted attribute adjacency graph with the subgraph isomorphism algorithm to achieve the feature recognition. Different from the above graph-based feature recognition methods, Vandenbrande and Requicha [8] put forward a trace-based method for recognizing features. Extracting all the feature traces from the geometric model of a part, the method adopts geometric reasoning to obtain the features of the part. Further, Gao and Shah [9] proposed a hybrid feature recognition approach which combines the graph-based feature recognition method with the trace-based one to gain the features of the part. In addition, Woo and Sakurai [10] and Yip-Hoi et al. [11] put forward a delta-volume decomposition feature recognition approach which mainly utilizes Boolean subtraction calculation between stock model and part model. Subsequently, Rameshbabu and Shunmugam [12] combined adjacency graph with volume subtraction techniques to recognize manufacturing features. Furthermore, Zhang et al. [13] came up with an ontology-based semantic approach to the recognition of machining features. Shi et al. [14] presented a heat kernel signature-based method for feature recognition. Although the above approaches are capable of feature recognition, they are not compatible with STEP-NC. To deal with the issue, many researchers did lots of significant research on

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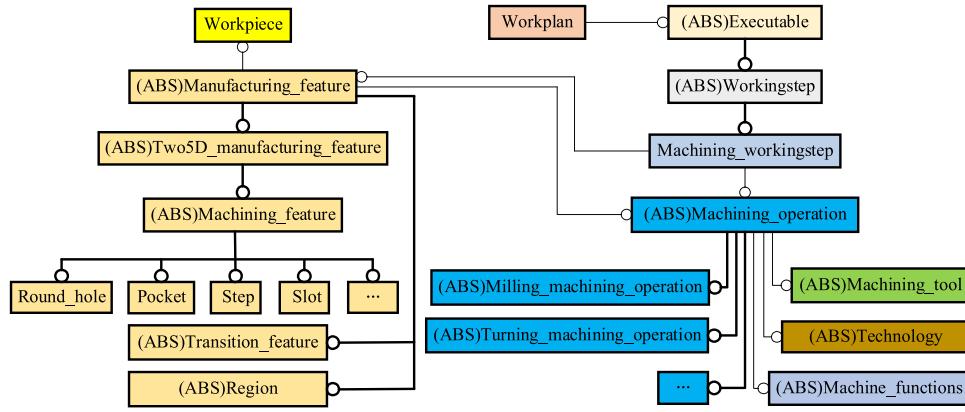


Fig. 1. The simplified data model of STEP-NC.

recognizing STEP-NC-compliant manufacturing features. For example, Wu and Han [15] put forward an attribute adjacency graph-based STEP-NC-compliant manufacturing features recognition method, which matches the feature subgraph by constructing attribute adjacency matrix. The method is time-consuming and inefficient because the subgraph matching is a non-deterministic polynomial problem. Subsequently, Xu et al. [16] combined graphs with rules to recognize the STEP-NC-compliant manufacturing features. Because of imperfect rules, the feature recognition had ambiguity and poor accuracy. Li and Chen [17] proposed a trace-based approach for STEP-NC-compliant manufacturing feature recognition. The approach is poor in versatility because the algorithm for the generation of feature traces depends on specific feature types. Later, Wang et al. [18] combined graphs with traces to recognize the STEP-NC-compliant intersection feature. The feature recognition is short of intelligence because the separation of intersection features is completed by manual. In addition, Guan et al. [19] proposed a STEP-NC-compliant manufacturing feature recognition method based on traditional neural network. Due to the shortcoming of traditional neural network, the accuracy of this method is low. Zhang et al. [20] realized the feature recognition compliant with STEP-NC using STEP-NC milling programs. **Because the generation of STEP-NC program is complex, it is inefficient.** Venu et al. [21] used JSDAI to develop a B-spline surface feature recognition system based on STEP. To sum up, although lots of significant research about feature recognition has been done, **most methods are either incompatible with STEP-NC**, or have the problems of time-consuming, low generality, poor intelligence and accuracy. To solve the above problems, this paper proposed an **artificial bee colony (ABC) algorithm** and **back propagation (BP) neural network-based intelligent feature** recognition method for **STEP-NC-compliant manufacturing**.

The remainder of this paper is organized as follows. Section 2 gives a brief introduction of STEP-NC data model. In Section 3, an intelligent feature recognition method for STEP-NC-compliant manufacturing features based on ABC algorithm and BP neural network is given. Section 4 demonstrates the effectiveness of the proposed method by case study. Conclusions and future works are drawn in Section 5.

2. STEP-NC data model

Other than ISO 6983, STEP-NC developed by ISO is a high-level and task-oriented NC machining data model, and is an extension of STEP in the field of numerical control. As shown Fig. 1, the workpiece consists of a series of manufacturing features, the workplan is composed of sequential workingsteps, and each manufacturing feature and each workingstep are closely linked together by the machining operation [22–24]. Therefore, the **object-oriented STEP-NC can completely and clearly describe all the information needed for machined parts**, and support two-way information flow between design and manufacture,

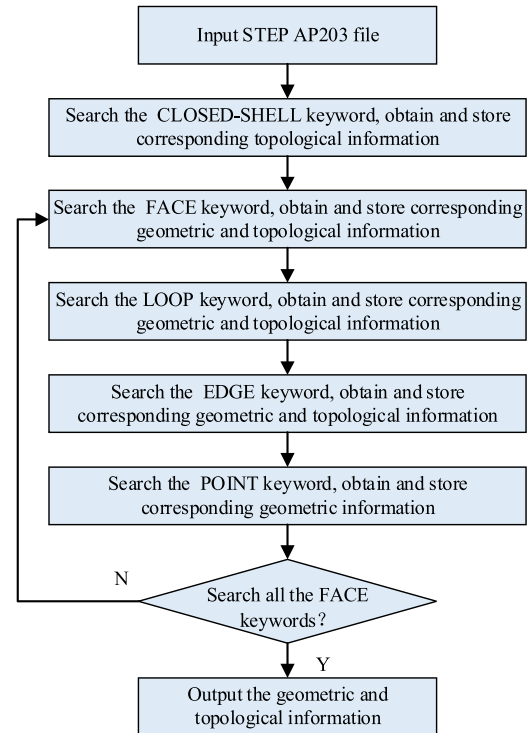


Fig. 2. The flow chart of geometric and topological information extraction based on STEP AP203.

which provides a condition for the realization of CAD/CAPP/CAM/CNC integration, even for intelligent manufacturing. In the actual production, STEP-NC 2.5D manufacturing features are most widely used. Thus, considering its practicability, **this paper mainly works on recognizing STEP-NC 2.5D manufacturing features.**

3. The method of intelligent feature recognition for STEP-NC-compliant manufacturing features

In order to solve the issues mentioned above, an intelligent feature recognition method for STEP-NC-compliant manufacturing features based on ABC algorithm and BP neural network is proposed in the section. The method is divided into two parts: constructing minimum subgraphs based on STEP AP203 and recognizing STEP-NC-compliant manufacturing features by ABC algorithm and BP neural network.

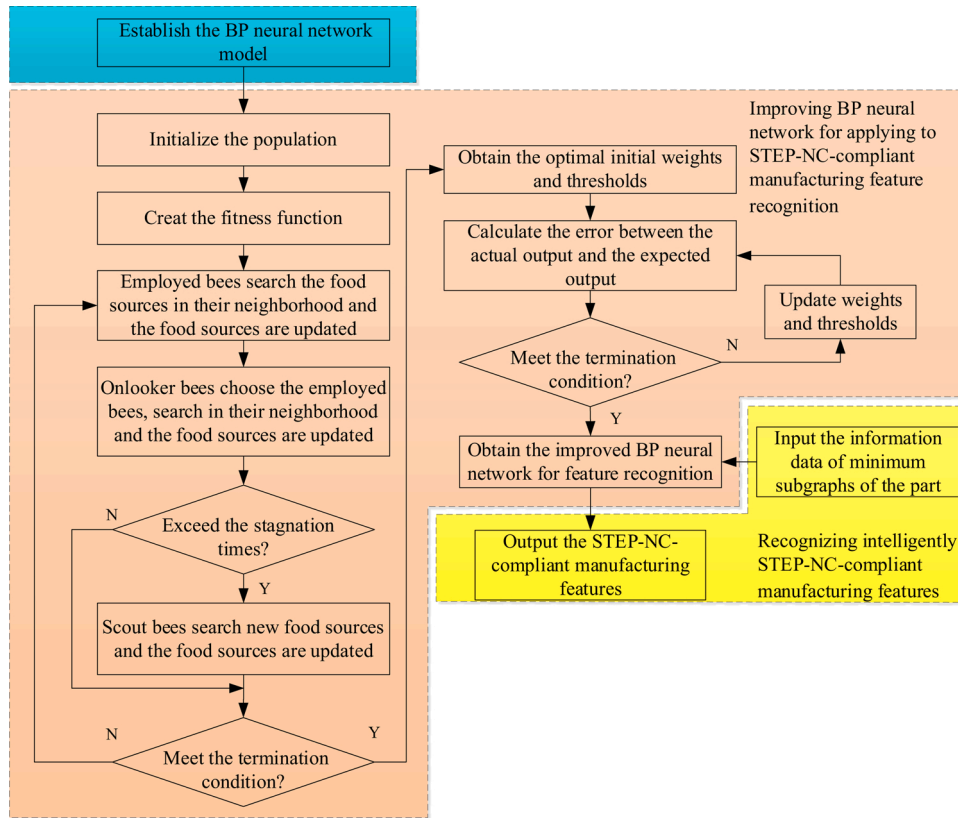


Fig. 3. The flow chart of STEP-NC-compliant manufacturing feature recognition based on ABC algorithm and BP neural network.

3.1. Constructing minimum subgraphs based on STEP AP 203

Because STEP-NC is an extension of STEP in the NC field, this paper extracts the geometric and topological information of the part from its STEP AP203 neutral file, further constructs its minimum subgraphs by judging the concavity and convexity of edges.

3.1.1. Extracting geometric and topological information based on STEP AP203

The main idea of extracting the geometric and topological information of the part from its STEP AP203 neutral file is to gradually search the CLOSED-SHELL, FACE, LOOP, EDGE and POINT keyword in the STEP AP203 physical file so as to obtain the required information. Further, the obtained information is stored in the database [25]. The flow chart is shown in Fig. 2.

3.1.2. Constructing minimum subgraphs by judging the concavity and convexity of edges

According to the extracted geometric and topological information of the part, the attribute adjacency graph of the part is established by judging the concavity and convexity of edges. Further, the minimum subgraphs of the part are generated. The main steps are described as follows: (1) The extracted faces are numbered and represented as the nodes. Moreover, the nodes are connected by the lines representing the intersecting edges between the faces; (2) Based on the concavity and convexity judgment algorithm [26], the concavity and convexity of the lines between the nodes is gained, which results in obtaining the attribute adjacency graph of the part; (3) After the convex lines and corresponding nodes are removed from the attribute adjacency graph of the part and the basal faces are chosen, the minimum subgraphs of the part are constructed.

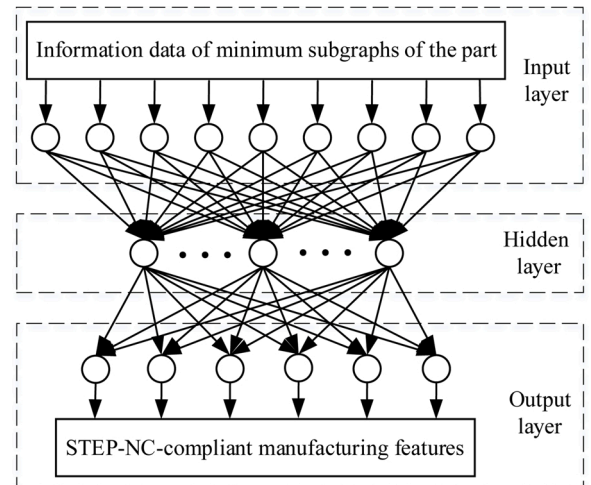


Fig. 4. The three-layer BP neural network model used for recognizing STEP-NC-compliant manufacturing features.

3.2. Recognizing STEP-NC-compliant manufacturing features by ABC algorithm and BP neural network

BP neural network [27,28] is a multi-layer feed-forward network trained by error back propagation algorithm, which has strong self-learning ability. However, it exists limitations such as slow convergence speed and local optimum and so on. ABC algorithm [29,30] is a swarm intelligence optimization algorithm that simulates the bee behavior gathering honey. It provides a new method for solving the efficient and global optimization problems. Therefore, this paper combines BP neural network and ABC algorithm to solve complex feature

Table 1

The weights of the edges and loops.

Concave edge	Convex edge	Concave loop	Convex loop
+1	-1	+5	-1

recognition problems. Based on it, an ABC algorithm and BP neural network-based STEP-NC-compliant manufacturing feature recognition method is proposed, which consists of establishing the BP neural network model for STEP-NC-compliant manufacturing feature recognition, improving BP neural network for applying to STEP-NC-compliant manufacturing feature recognition and recognizing intelligently STEP-NC-compliant manufacturing features, as shown in Fig. 3.

3.2.1. Establishing the BP neural network model for STEP-NC-compliant manufacturing feature recognition

The three-layer BP neural network model for STEP-NC-compliant manufacturing feature recognition is established and shown in Fig. 4.

(1) Input layer:

The input of the established BP neural network is the information data of minimum subgraphs of the part obtained above, i.e. the score of each face in the minimum subgraphs. The score G of each face can be calculated by:

$$G = \sum_{i=1}^m E_i + \sum_{j=1}^n L_j \quad (1)$$

where E_i is the weight of the i^{th} side of the face, L_j is the weight of the j^{th} loop of the face, which are determined according to Table 1 [31]; m is the number of edges, n is the number of loops.

(2) Hidden layer:

The number of nodes in the hidden layer is determined by Hecht-Nelson method [32]. According to it, the number n_2 of nodes in the hidden layer is represented as $n_2 = 2n_1 + 1$, where n_1 is the number of input nodes.

(3) Output layer:

The output of the established BP neural network is STEP-NC-compliant manufacturing feature. For the purpose of practicality, six common STEP-NC 2.5D manufacturing features, i.e. slot, open pocket, step, round hole, closed pocket and boss are considered in the paper, whose output can be represented as [1, 0, 0, 0, 0, 0], [0, 1, 0, 0, 0, 0], [0, 0, 0, 1, 0, 0], [0, 0, 0, 1, 0, 0], [0, 0, 0, 0, 1, 0] and [0, 0, 0, 0, 0, 1], respectively.

3.2.2. Improving BP neural network for applying to STEP-NC-compliant manufacturing feature recognition

In order to improve the accuracy and efficiency of feature recognition, an improved BP neural network combining with ABC algorithm is proposed. Its key steps are as follows:

(2) Population initialization:

The initial population contains S_N food sources, which is equal to the number of employed bees and the number of onlooker bees. Every food source is looked as a solution which consists of the weight matrix w_{ij} between the input layer and the hidden layer, the weight matrix w_{jk} between the hidden layer and the output layer, the threshold matrix a_j of the hidden layer, and the threshold matrix b_k of the output layer. And each solution is a D-dimensional which can be obtained by:

$$D = N_{\text{input}} \times N_{\text{hidden}} + N_{\text{hidden}} + N_{\text{hidden}} \times N_{\text{output}} + N_{\text{output}} \quad (2)$$

where N_{input} , N_{hidden} , and N_{output} are the number of neurons in the

input layer, the hidden layer and the output layer in the BP neural network, respectively.

(3) The creation of fitness function:

The fitness function is an index to evaluate the merits of an individual and can accurately reflect the merits of a solution. Because the output H_j of the hidden layer in the BP neural network can be calculated by:

$$H_j = f_1 \left(\sum_{i=1}^{n_1} \omega_{ij} \times y_i - a_j \right) \quad j = 1, 2, \dots, n_2 \quad (3)$$

where n_1 is the number of input layers, n_2 is the number of hidden layers, y_i is the input of the BP neural network, $f_1(x)$ is the transfer function between the input layer and the hidden layer.

The output O_k of the output layer in the BP neural network can be calculated by:

$$O_k = f_2 \left(\sum_{j=1}^{n_2} H_j \omega_{jk} - b_k \right) \quad k = 1, 2, 3, \dots, n_3 \quad (4)$$

where n_3 is the number of output layers, $f_2(x)$ is the transfer function between the hidden layer and the output layer.

Thus, the mean square error (MSE) can be calculated by:

$$MSE = \frac{1}{p \times n_3} \sum_{p=1}^p \sum_{k=1}^{n_3} \left(d_k^{(p)} - O_k^{(p)} \right)^2 \quad (5)$$

where p is the number of samples, d_k is the expected output.

Based on the above, the fitness function $f(X_i)$ is created as:

$$f(X_i) = \begin{cases} 1 & MSE_i = 0 \\ \frac{1}{1 + MSE_i} & MSE_i > 0 \end{cases} \quad (6)$$

where MSE_i is the mean square error of the i^{th} solution X_i , $i = 1, 2, 3, \dots, S_N$.

(4) The neighborhood search of employed bees:

The employed bees carry out the neighborhood search according to Eq. (7), and the fitness of the new solution is calculated according to Eq. (6). If the fitness of the new solution is better than the fitness of the old one, the old solution is replaced by the new one.

$$V_{ij} = X_{ij} + \text{rand}(-1, 1)(X_{ij} - X_{kj}) \quad (7)$$

where V_{ij} is the j^{th} component of the i^{th} new solution, X_{ij} is the j^{th} component of the i^{th} old solution, X_{kj} is the j^{th} component of the k^{th} old solution, $j \in \{1, 2, \dots, D\}$, $k \in \{1, 2, \dots, S_N\}$, $k \neq i$.

(5) The neighborhood search of onlooker bees:

According to the Eq. (8), the onlooker bees select the employed bees by the roulette method. Based on it, the onlooker bees carry out the neighborhood search according to Eq. (7), and the fitness of the new solution is calculated according to Eq. (6). If the fitness of the new solution is better than the fitness of the old one, the old solution is replaced by the new one.

$$P_i = \frac{f(X_i)}{\sum_{n=1}^{S_N} f(X_n)} \quad (8)$$

where P_i is the selection probability of the i^{th} solution.

(6) The judgment of stagnation times:

If the stagnation times of the solution exceed the

Table 2

The input and desired output for each feature in the sample database.

Feature	Input	Desired output
Step	-3,-3, 0, 0, 0, 0, 0, 0	0, 0, 1, 0, 0, 0
Boss	1,-3, -3,-3,-3, 0, 0, 0	0, 0, 0, 0, 0, 1
Step	-3,-7, 0, 0, 0, 0, 0, 0	0, 0, 1, 0, 0, 0
Closed pocket	18,1,1,1,1, -3, -3, -3, -3	0, 0, 0, 0, 1, 0
Slot	-1,-3,-3, 0, 0, 0, 0, 0	1, 0, 0, 0, 0, 0
...	...	...
Closed pocket	6,-3,-3,-3,-3, 0, 0, 0	0, 0, 0, 0, 1, 0
Step	-3,-3, 0, 0, 0, 0, 0, 0	0, 0, 1, 0, 0, 0
Open pocket	1,-1,-1,-2, 0, 0, 0, 0	0, 1, 0, 0, 0, 0
Round hole	-1, -4, 0, 0, 0, 0, 0, 0	0, 0, 0, 1, 0, 0
Open pocket	1, 1,-1,-1, 0, 0, 0, 0	0, 1, 0, 0, 0, 0
Round hole	-1, -1, 0, 0, 0, 0, 0, 0	0, 0, 0, 1, 0, 0
Slot	1, -3, -3, 0, 0, 0, 0, 0	1, 0, 0, 0, 0, 0

predetermined maximum times, the scout bees will search the new solution according to Eq. (9), and the update of the solution is implemented as said above. Otherwise, go to the step (6).

$$X_{ij} = X_{min,j} + rand(0, 1)(X_{max,j} - X_{min,j}) \quad (9)$$

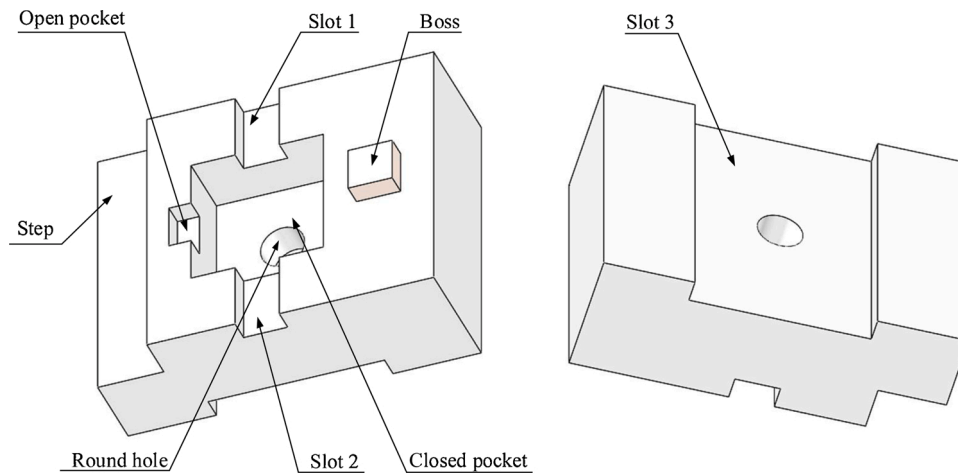
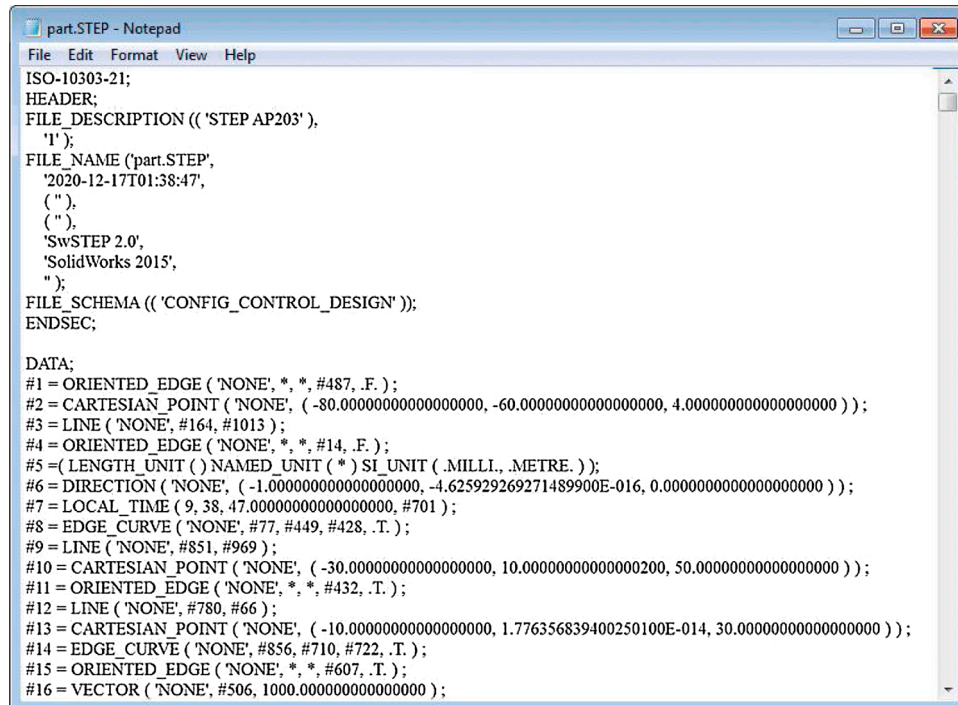
where $X_{min,j}$ is the j^{th} component of the minimum solution, $X_{max,j}$ is the j^{th} component of the maximum solution.

(7) The judgment of the termination condition:

If the termination condition is satisfied, the initial optimal weights and thresholds are obtained and go to the step (7); if not, return the step (3).

(8) The training of the improved BP neural network:

Based on the initial optimal weights and thresholds of the BP neural network obtained above and the data in Table 2, the improved BP neural network is trained by using the S-type tangent function $\text{tansig}()$, S-type logarithmic function $\text{logsig}()$

**Fig. 5.** The 3D model of a part.**Fig. 6.** The STEP AP203 file of the part.

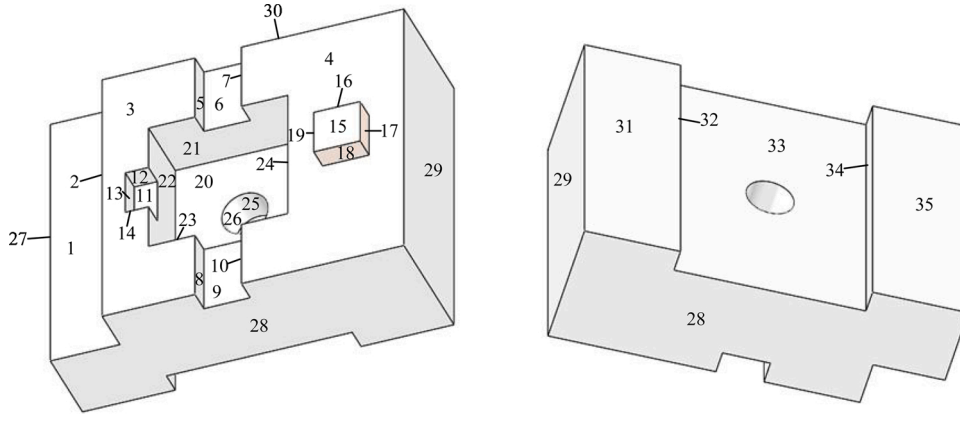


Fig. 7. The extracted faces and their number based on STEP AP203.

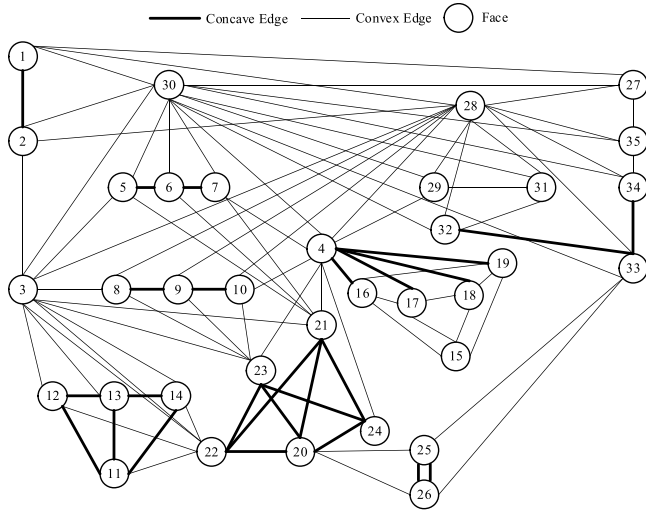


Fig. 8. The attribute adjacency graph of the part.

and trainidx algorithm, and setting the training times as 500, the training error target as 0.001 and the learning rate as 0.01 [33]. When the training termination condition is satisfied, the improved BP neural network for STEP-NC-compliant manufacturing feature recognition is obtained.

3.2.3. Recognizing intelligently STEP-NC-compliant manufacturing features

After the information data of the minimum subgraphs of the part to be recognized is input to the trained BP neural network, its features can be intelligently achieved by comparing the actual outputs with the expected outputs.

4. Case study

The 3D model of a part containing the slot, boss, open pocket, step, round hole and closed pocket features and its STEP AP203 neutral file are given in Figs. 5 and 6, respectively. Based on the method proposed in this paper, the intelligent feature recognition of the part is realized by using MATLAB. The relevant parameters [34] are set as follows: the number of food sources is 50, the maximal stagnation times are 10, and the maximum iteration times are 500.

Firstly, the geometric and topological information of the part is extracted based on its STEP AP203 file, and the extracted faces are numbered, as shown in Fig. 7.

Then, the attribute adjacency graph and the minimum subgraphs of the part are constructed according to the concavity and convexity

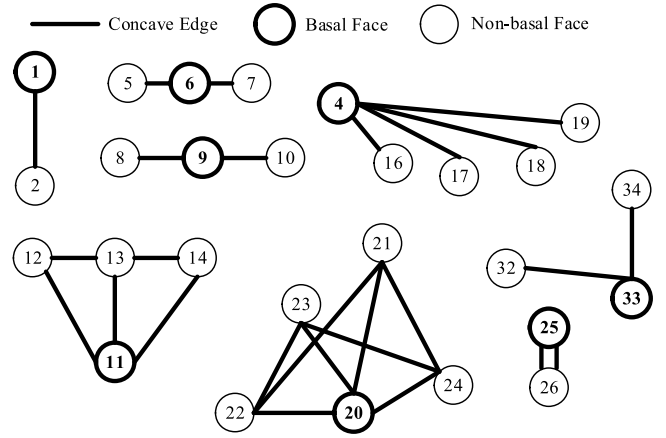


Fig. 9. The minimum subgraphs of the part.

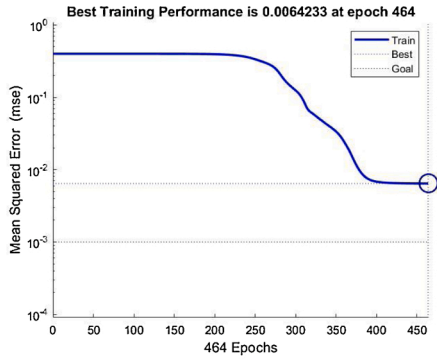
Table 3

The input and desired output for each feature in the part.

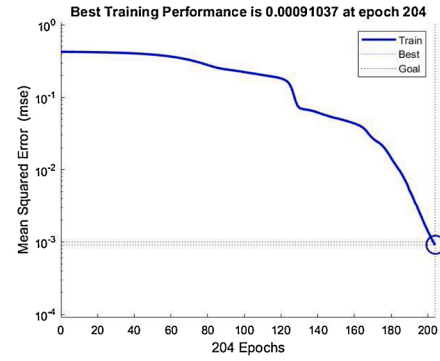
Feature	Input	Desired output
Slot 1	-1,-3,-3, 0, 0, 0, 0, 0	1, 0, 0, 0, 0, 0
Slot 2	-1,-3,-3, 0, 0, 0, 0, 0	1, 0, 0, 0, 0, 0
Step	-3,-3, 0, 0, 0, 0, 0, 0	0, 0, 1, 0, 0, 0
Round hole	-1, -1, 0, 0, 0, 0, 0, 0	0, 0, 0, 1, 0, 0
Open pocket	1, 1,-1,-1, 0, 0, 0, 0	0, 1, 0, 0, 0, 0
Closed pocket	6, 1,-3,-3,-3, 0, 0, 0	0, 0, 0, 0, 1, 0
Boss	0,-3,-3,-3,-3, 0, 0, 0	0, 0, 0, 0, 0, 1
Slot 3	-4,-3,-3, 0, 0, 0, 0, 0	1, 0, 0, 0, 0, 0

judgment algorithm, as shown in Figs. 8 and 9. Further, the input and desired output for each feature in the part are obtained, as shown in Table 3.

Finally, the improved BP neural network is used to recognize the features of the part. To show the performance of the improved BP neural network, the traditional BP neural network and the improved BP neural network proposed in this paper are firstly trained based on the information in Table 2. The training error convergence curves of two algorithms are gained by MATLAB, as shown in Fig. 10. It shows the traditional BP neural network falls into the local optimum when it is trained to 464 times, while the BP neural network based on ABC algorithm gains global optimal when it is trained to 204 times. Also, the prediction errors of the two algorithms are 0.9754 and 0.2011, respectively. It can be seen that the performance of the improved BP neural network is better. Further, for showing its advantages in feature recognition, Table 4 gives the results of feature recognition for the proposed



(a) The training error convergence curve of traditional BP neural network



(b) The training error convergence curve of the improved BP neural network

Fig. 10. The training error convergence curves of two algorithms.

Table 4

The results of feature recognition of two methods.

Feature	Actual output from the method in Reference [19]	Result	Actual output from the method in this paper	Result
Slot 1	0.9431,0.1064,-0.0655, 0.0165,-0.0245,-0.0499	Slot	0.9752,0.0043,0.0199, 0.0004,0.0003,0.0003	Slot
Slot 2	0.9431,0.1064,-0.0655, 0.0165,-0.0245,-0.0499	Slot	0.9752,0.0043,0.0199, 0.0004,0.0003,0.0003	Slot
Step	-0.1230,-0.0781,1.0674, 0.0809,0.0849,-0.1163	Step	0.0102,0.0070,0.9922, 0.0065,0.0001,0.0210	Step
Round hole	0.0111,0.0636,-0.0909, 0.0322,0.9020,-0.1680	Round hole	0.0007,0.1129,0.0016, 0.9984,0.0000,0.0006	Round hole
Open pocket	0.0013,-0.0294,0.9249, -0.0180,0.0091,0.0373	Step	0.0000,0.9887,0.0180, 0.0029,0.0019,0.0040	Open pocket
Closed pocket	-0.0072,-0.0257,0.0303, 0.0054,-0.0992,1.0435	Closed pocket	0.0006,0.1847,0.0032, 0.0000,0.9874,0.0000	Closed pocket
Boss	0.9503,0.0429,0.0221, 0.9914,-0.0038,0.1006	N/A	0.0020,0.0076,0.0009, 0.0091,0.0016,0.9998	Boss
Slot 3	0.9291,0.0096,0.0128, 0.0069,0.0308,0.1254	Slot	0.9774,0.0268,0.0504, 0.0112,0.0001,0.0028	Slot

method in this paper and the method presented in Reference [19]. It can be seen from Table 4 that the open pocket is judged wrong to be the step and the boss is not recognized by the method in Reference [19], while all the recognized features are correct by the intelligent feature recognition method proposed in this paper. Moreover, in the same computer environment (2.70 GHz CPU, 4GB memory and Windows 7 operating system), the feature recognition by the method in Reference [19] takes 0.06 s, while the feature recognition by the intelligent feature recognition method proposed in this paper takes 0.012 s.

5. Conclusion

For solving the existing problems of feature recognition at present, an ABC algorithm and BP neural network-based intelligent feature recognition method for STEP-NC-compliant manufacturing features was come up with in this paper. In the method, the use of STEP AP 203 file realizes seamless connection between CAD and feature recognition; the establishment of the three-layer BP neural network model for STEP-NC-compliant manufacturing feature recognition lays a foundation for intelligently identifying features. Because the model has strong adaptability and extensibility, the generality of the feature recognition is enhanced; the combination of ABC algorithm with BP neural network achieves accurate and efficient feature recognition for STEP-NC-compliant manufacturing features, which not only provides an effective method to recognize STEP-NC-compliant manufacturing feature intelligently, accurately and efficiently, but also provides a powerful technical support for the integration of CAD/CAPP/CAM/CNC, even for the realization of intelligent manufacturing. In the future, the feature recognition for the STEP-NC-compliant transition features and freeform surface features will be studied.

Declaration of Competing Interest

The authors declare that there is no conflict of interest.

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References

- [1] ISO 10303-1: Part 203. Industrial automation systems and integration-product data representation and exchange-part 203: application protocol: configuration controlled 3D design of mechanical parts and assemblies. 2011.
- [2] Kadir AA, Xu X. Towards high-fidelity machining simulation. *J Manuf Syst* 2011;30(3):175–86.
- [3] Rauch M, Laguionie R, Hascoet JY, Suh SH. An advanced STEP-NC controller for intelligent machining processes. *Robot Comput-Integr Manuf* 2012;28(3):375–84.
- [4] Hardwick M, Zhao YF, Proctor FM, Nassehi A, Xu X, Venkatesh S, et al. A roadmap for STEP-NC-enabled interoperable manufacturing. *Int J Adv Manuf Technol* 2013; 68(5–8):1023–37.
- [5] Joshi S, Chang TC. Graph-based heuristics for recognition of machined features from a 3D solid model. *Comput Aided Des* 1988;20(2):58–66.
- [6] Sunil VB, Agarwal R, Pande SS. An approach to recognize interacting features from B-rep CAD models of prismatic machined parts using a hybrid (graph and rule based) technique. *Comput Ind* 2010;61(7):686–701.
- [7] Xie F, Guo Y, Zhang HL, Wang FL, Huang X. Manufacturing feature recognition based on graph and subgraph isomorphism algorithm. *J Nanjing Univ Aeronaut Astronaut* 2018;50(3):390–6.
- [8] Vandenbrande JH, Requicha AAG. Spatial reasoning for the automatic recognition of machinable features in solid models. *IEEE Trans Pattern Anal Mach Intell* 1993; 15(12):1269–85.
- [9] Gao S, Shah JJ. Automatic recognition of interacting machining features based on minimal condition subgraph. *Comput Aided Des* 1998;30(9):727–39.
- [10] Woo Y, Sakurai H. Recognition of maximal features by volume decomposition. *Comput Aided Des* 2002;34(3):195–207.
- [11] Yip-Hoi D, Dutta D, Huang ZD. A customizable machining feature extraction methodology for turned components. *J Manuf Syst* 2003;22(2):82–98.
- [12] Rameshbabu V, Shunmugam MS. Hybrid feature recognition method for setup planning from STEP AP-203. *Robot Comput-Integr Manuf* 2009;25(2):393–408.
- [13] Zhang YZ, Luo XF, Zhang BY, Zhang SH. Semantic approach to the automatic recognition of machining features. *Int J Adv Manuf Technol* 2017;89(1–4):417–37.

- [14] Shi Y, Zhang YC, Baek S, De Backer W, Harik R. Manufacturability analysis for additive manufacturing using a novel feature recognition technique. *Comput Aided Des Appl* 2018;15(6):941–52.
- [15] Wu XD, Han ZX. Implementation of STEP-NC manufacturing feature recognition based on attribute adjacent graph. *Mach Des Manuf Eng* 2013;42(7):13–6.
- [16] Xu SX, Anwer N, Qiao LH. Feature recognition for virtual machining. 21st International Conference on Industrial Engineering and Engineering Management 2014:123–7.
- [17] Li MZ, Chen R. A study of feature recognition based on hint from STEP files. *J Shanxi Univ Sci Technol* 2011;29(4):57–62.
- [18] Wang J, Ou DJ, Shu QL, Wang GX. Interacting feature recognition technology based on STEP-NC. *Comput Integr Manuf Syst* 2014;20(5):1051–61.
- [19] Guan XS, Meng GW, Yuan XH. Machining feature recognition of part from STEP file based on ANN. 2010 International Conference on Computer, Mechatronics, Control and Electronic Engineering 2010:54–7.
- [20] Zhang XZ, Nassehi A, Newman ST. Feature recognition from CNC part programs for milling operations. *Int J Adv Manuf Technol* 2014;70(1–4):397–412.
- [21] Venu B, Komma VR, Srivastava D. STEP-based feature recognition system for B-spline surface features. *Int J Autom Comput* 2018;15(4):500–12.
- [22] Zhang Y, Xu X, Liu YX, Bai XL. Service-oriented, cross-platform and high-level machining simulation. *Int J Comput Integ Manuf* 2012;25(3):280–95.
- [23] ISO 14649: Part 10. Industrial automation systems and integration physical device control-data model for computerized numerical controllers–part 10: general process data. 2004.
- [24] Xu XW, Wang H, Mao J, Newman ST, Kramer TR, Proctor FM, et al. STEP-compliant NC research: the search for intelligent CAD/CAPP/CAM/CNC integration. *Int J Prod Res* 2005;43(17):3703–43.
- [25] Salonitis K, Stavropoulos P. On the integration of the CAX systems towards sustainable production. *Procedia CIRP* 2013;9:115–20.
- [26] Tao JH, Yang XQ, Liu XC, Zhu MY. Research of NC automatic programming based on technological feature recognition. *Comput Eng Des* 2011;32(10):3548–52.
- [27] Liu Y. An improved AHP and BP neural network method for service quality evaluation of city bus. *Int J Comput Appl Technol* 2018;58(1):37–44.
- [28] Amin S. Backpropagation-artificial neural network (BP-ANN): understanding gender characteristics of older driver accidents in West Midlands of United Kingdom. *Saf Sci* 2020;122. 104539-1-104539-11.
- [29] Kalayci CB, Hancilar A, Gungor A, Gupta SM. Multi-objective fuzzy disassembly line balancing using a hybrid discrete artificial bee colony algorithm. *J Manuf Syst* 2015;37(3):672–82.
- [30] Prasanth RSS, Hans Raj K. Optimization of straight cylindrical turning using artificial bee colony (ABC) algorithm. *J Ins Eng Ser C* 2017;98(2):171–7.
- [31] Jian CF, Li M, Qiu KY, Zhang MY. An improved NBA-based STEP design intention feature recognition. *Future Gener Comput Syst* 2018;88:357–62.
- [32] Zhang YX, Gao XD, Katayama S. Weld appearance prediction with BP neural network improved by genetic algorithm during disk laser welding. *J Manuf Syst* 2015;34:53–9.
- [33] Sunil VB, Pande SS. Automatic recognition of machining features using artificial neural networks. *Int J Adv Manuf Technol* 2009;41(9–10):932–47.
- [34] Duan LT, Han DZ, Tian QT. Design of intrusion detection system based on improved ABC_elite and BP neural networks. *Comput Sci Inf Syst* 2019;16(3): 773–95.